Ericsson PE Node IND-PE-03 Indiranagar Exchange Hardware/Environmental Fault Troubleshooting and Resolution SOP

Node: IND-PE-03 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_3070

Description

The Ericsson PE Node IND-PE-03 Indiranagar Exchange Hardware/Environmental Fault failure domain occurs due to the inherent vulnerability of the ASIC memory parity error and line card crash mechanisms in the PE role. The PE role is responsible for the aggregation and forwarding of user traffic, and any hardware or environmental faults in this domain can have a cascading impact on the overall network reliability and performance. The loss of power supply redundancy in the PE node can also lead to a sudden failure of the node, resulting in a warning alarm (ALARM_3070) being raised. The underlying protocol mechanics of the Ericsson equipment, specifically the VPLS (Virtual Private LAN Service) and the LDP (Label Distribution Protocol), rely on the correct functioning of the ASIC memory and line cards to maintain the integrity of the control and data planes. Therefore, any faults in this domain can have a significant impact on the network's ability to provide reliable and efficient services.

Pre-requisite Checks

- pre_requisite: Verify the Indiranagar Exchange site has been locked down and access restricted to authorized personnel.

- pre_requisite: Check the redundant link status to ensure it is operational and ready for switchover in case of a failure.

- pre_requisite: Dispatch remote hands to the Indiranagar Exchange site to assist with the troubleshooting and resolution process.

Potential Root Causes

Cause	Probability
ASIC memory parity error due to manufacturing defect or aging	High
Line card crash due to software bug or hardware failure	Medium
Power supply redundancy loss due to hardware failure or power cabling issue	Low

Diagnostic Commands & Expected Outputs

```
Command: show system redundancy status, Expected Output: Redundancy Status: Active/Active, Description: This command checks the current redundancy status of the PE node, ensuring that both power supplies are operational and ready for switchover.
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```
Command: show system environment temperature, Expected Output: Temperature: 25°C, Description: This command checks the current temperature of the PE node, ensuring that it is within the acceptable operating range.

Command: show system asic memory status, Expected Output: ASIC Memory Status: Normal, Description: This command checks the current status of the ASIC memory, ensuring that it is functioning correctly and not experiencing any errors.
```

Resolution Steps

Step 1: Step: Replace the faulty power supply unit, Command: system redundancy replace power-supply , Description: Replace the faulty power supply unit with a new one, ensuring that the redundancy is maintained and the PE node remains operational.

Step 2: Step: Replace the faulty line card, Command: system line-card replace , Description: Replace the faulty line card with a new one, ensuring that the PE node remains operational and the user traffic is not affected.

Step 3: Step: Replace the faulty ASIC, Command: system asic replace , Description: Replace the faulty ASIC with a new one, ensuring that the PE node remains operational and the user traffic is not affected.

Post-Check Verification

```
Command: show system redundancy status, Expected Output: Redundancy Status: Active/Active, Description: Verify that the redundancy status of the PE node is Active/Active, ensuring that both power supplies are operational and ready for switchover.

Command: show system environment temperature, Expected Output: Temperature: 25°C, Description: Verify that the current temperature of the PE node is within the acceptable operating range.

Command: show system asic memory status, Expected Output: ASIC Memory Status: Normal, Description: Verify that the current status of the ASIC memory is functioning correctly and not experiencing any errors.
```

Roll-back Procedures

- Step: Restore the original power supply unit, Command: system redundancy restore power-supply , Description: Restore the original power supply unit, ensuring that the redundancy is maintained and the PE node remains operational.

- Step: Restore the original line card, Command: system line-card restore , Description: Restore the original line card, ensuring that the PE node remains operational and the user traffic is not affected.

- Step: Restore the original ASIC, Command: system asic restore , Description: Restore the original ASIC, ensuring that the PE node remains operational and the user traffic is not affected.